

11.1 Atoms, Nuclei & Radiation

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.1 Atoms, Nuclei & Radiation
Difficulty	Medium

Time allowed: 20

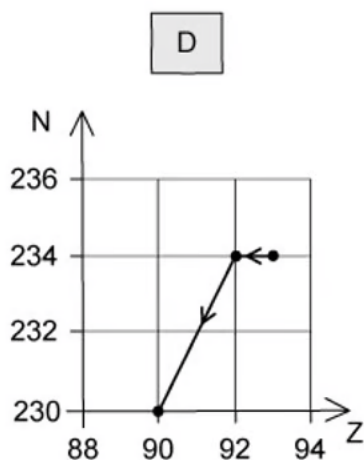
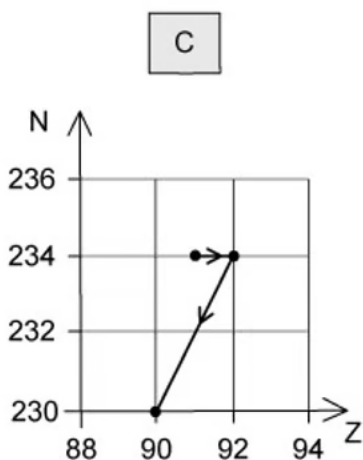
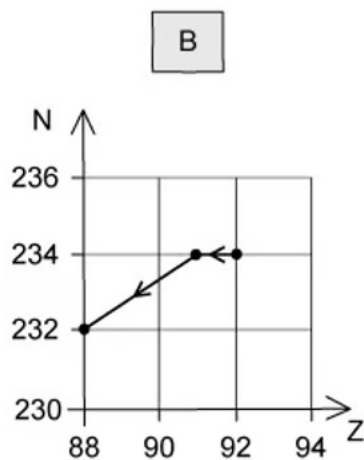
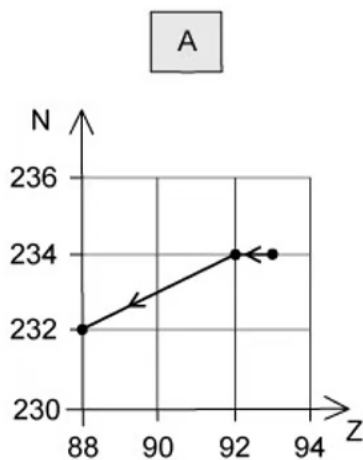
Score: /15

Percentage: /100

Question 1

A radioactive nucleus is formed by β -decay. This nucleus then decays by α -emission.

The graphs below show the nucleon number N plotted against proton number Z . Which one shows the β -decay followed by the α -emission?



[1 mark]

Question 2

When α -particles are directed at gold leaf

- 1 almost all α -particles pass through without deflection,
- 2 a few α -particles deviate through large angles.

What are the reasons for these effects?

	1	2
A	the positive charge in an atom is not concentrated enough to deflect an α -particle	occasionally an α -particle experiences many small deflections in the same direction
B	the gold nucleus is very small so most α -particles miss all nuclei	occasionally the path of an α -particle is close to a nucleus
C	most α -particles miss all gold atoms	a few α -particles bounce off gold atoms
D	most α -particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α -particles bounce back from the gold surface

[1 mark]

Question 3

Nucleus X decays in two stages to produce nucleus Y.

Which decay sequence will result in the highest number of neutrons in nucleus Y?

- A. an α -particle followed by a β -particle
- B. a β -particle followed by a γ -ray
- C. a β -particle followed by another β -particle
- D. an α -particle followed by a γ -ray

[1 mark]

Question 4

The isotope $^{222}_{86}\text{Rn}$ decays in a sequence of emissions to form the isotope $^{206}_{82}\text{Pb}$.

It will either emit an α -particle or a β -particle at each stage of the decay sequence.

What is the number of stages in the decay sequence?

- A. 20
- B. 16
- C. 8
- D. 4

[1 mark]

Question 5

An element emits an alpha particle from its radioactive nucleus.

The daughter nucleus then emits a beta particle, and then the daughter nucleus of that reaction emits another beta particle.

Which statement describes the final nuclide that is formed?

- A. it is a nuclide of the same element but with different proton number
- B. it is a nuclide of a different element of higher proton number
- C. it is a different isotope of the original element
- D. it is identical to the original nuclide

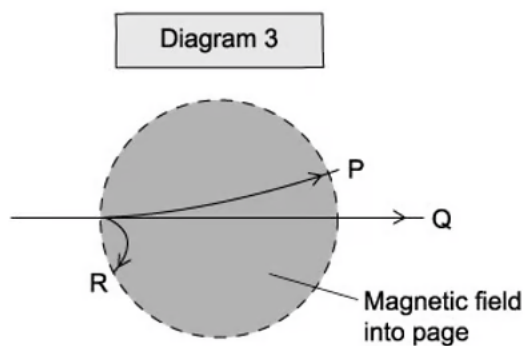
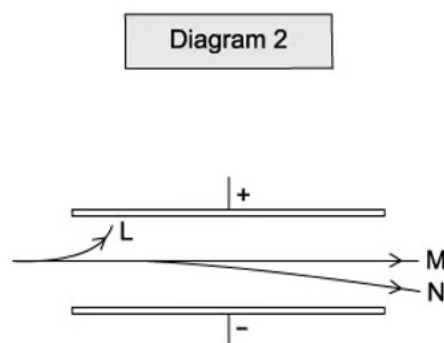
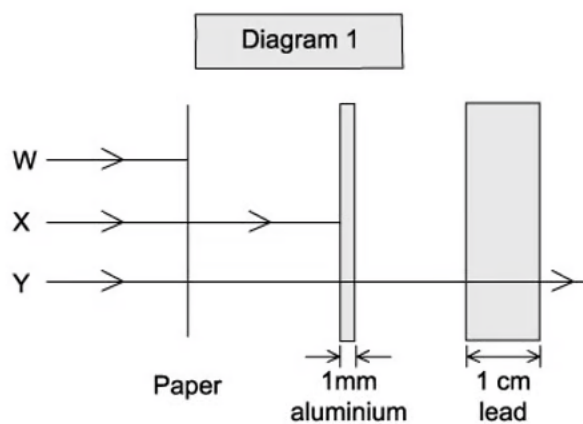
[1 mark]

Question 6

Alpha, beta and gamma radiations:

- 1 are absorbed to different extents in solids
- 2 behave differently in an electric field
- 3 behave differently in a magnetic field

Diagrams 1, 2 and 3 illustrate these behaviours .



Which three labels on these diagrams refer to the **same** kind of radiation?

- A. X, L, R
- B. W, L, R

C. W, L, P

D. Y, M, P

[1 mark]

Question 7

The nuclides shown in the grid below are arranged according to the number of protons and neutrons in each.

A nucleus of the nuclide ${}^8_3\text{Li}$ decays by emitting a β -particle.

What is the resulting nuclide?

Number of protons	4					D	C	
	3				${}^6_3\text{Li}$	${}^7_3\text{Li}$	${}^8_3\text{Li}$	
	2		${}^3_2\text{He}$	${}^4_2\text{He}$			B	A
	1	${}^1_1\text{H}$	${}^2_1\text{H}$					
		0	1	2	3	4	5	6
		Number of neutrons						

[1 mark]

Question 8

A thorium isotope has a nucleon number of 232 and a proton number of 90. It decays to form another isotope of with a nucleon number of 228.

How many alpha particles and beta particles are emitted during this decay?

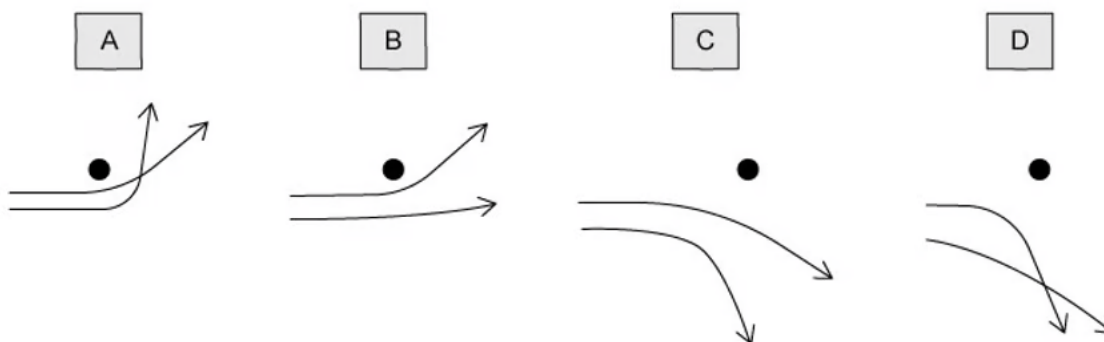
	Alpha particles	Beta particles
A	0	4
B	1	2
C	1	1
D	2	1

[1 mark]

Question 9

Two α -particles with equal energies are deflected by a gold nucleus.

Which diagram best represents their paths ?

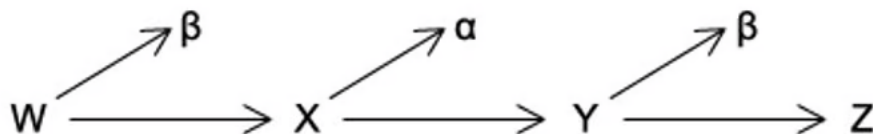


[1 mark]

Question 10

Three successive radioactive decays are shown in the diagram below; each one results in a particle being emitted.

The first decay results in the emission of a β -particle. The second decay results in the emission of an α -particle. The third decay results in the emission of another β -particle.



Nuclides W and Z are compared. Which statement is correct?

- A. W and Z are isotopes of the same element
- B. Z is a different element of reduced mass
- C. Z is a different element of lower atomic number
- D. W and Z are identical in all respects

[1 mark]

Question 11

Antimatter is a particle that is an antiparticle to the corresponding particle. A positron is the antiparticle of an electron.

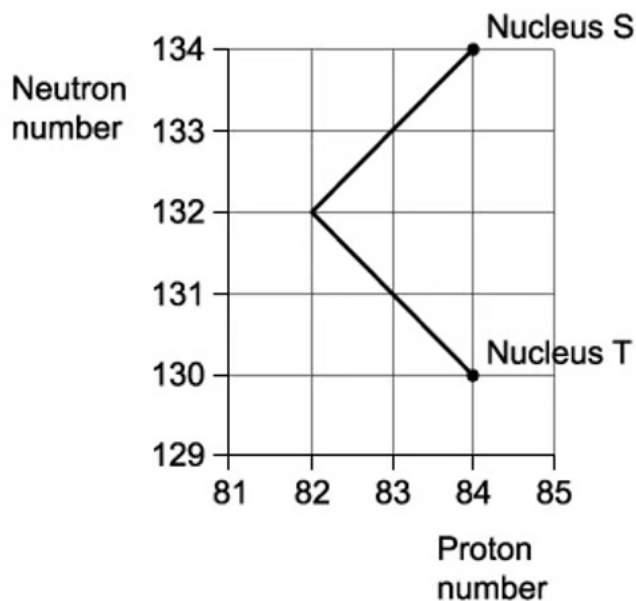
What is the difference between a positron and an electron?

- A. Mass
- B. magnitude of charge
- C. Charge
- D. Spin

[1 mark]

Question 12

A sequence of radioactive decays is shown in the graph of neutron number against proton number.



Nucleus S is at the start of the sequence and, after the decays have occurred, nucleus T is formed.

What is emitted during the sequence of decays?

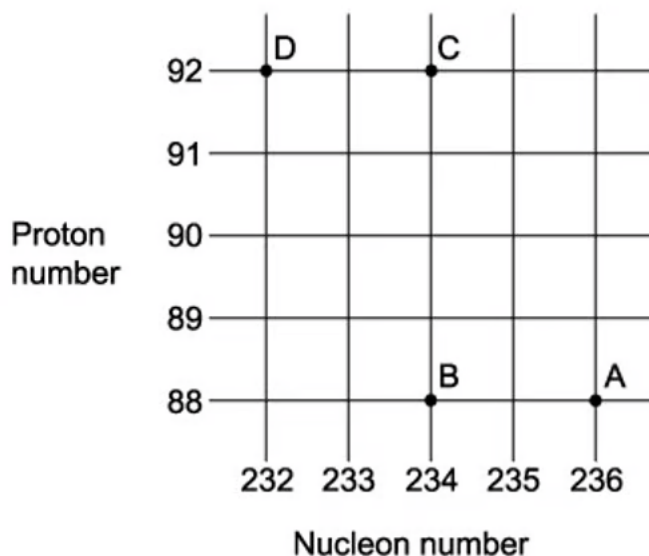
- A. one α -particle followed by one β -particle
- B. two β -particles followed by one α -particle
- C. two α -particles followed by two β -particles
- D. one α -particle followed by two β -particles

[1 mark]

Question 13

A radioactive substance with a nucleon number of 234 and a proton number of 90, decays by β -emission into a daughter product which in turn decays by further β -emission into a granddaughter product.

Which letter in the diagram represents the granddaughter product?



[1 mark]

Question 14

Astatine is a radioactive substance; it has a nucleon number of 218 and a proton number of 85. When it decays it forms a polonium nucleus, emitting a β^- particle and an α -particle.

What are the nucleon number and the proton number of the polonium nucleus?

	Nucleon number	Proton number
A	214	84
B	214	83
C	216	83
D	215	82

[1 mark]

Question 15

An element with an unstable nucleus decays by emitting an alpha particle to become the nucleus of a different element.

The nucleus of the new element is unstable and will emit either an α -particle or a β^- particle. This process continues until an isotope of the original element is formed.

What is the minimum possible number of the particles emitted?

- A. 5
- B. 4
- C. 3
- D. 2

[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}^0_{-1}\beta + \bar{\nu}_e$
 - The mass is not changed, but the proton number increases by one; so the Z number has increased from 91 to 92
- An α -particle is made of two protons and two neutrons,
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- The nucleon number is reduced by 4 making this 230
- The proton number is reduced by 2 making this 90
 - Graph **C** is therefore the correct option

2

The correct answer is **B** because:

- The gold foil experiment directed parallel beams of α -particles at a piece of gold foil only 10^{-6} m thick
- Most of the α -particles went straight through the foil
 - This is due to the small size of the nucleus of the atom
- Some of the α -particles were deflected through large angles
 - This is due to the path occasionally being close to the nucleus

A is incorrect because alpha particles are deflected by the positive charge when it is close enough to the nucleus

C is incorrect because the alpha particles did not bounce off the nuclei but were deflected due to repulsion from positively charged protons

D is incorrect because the gold foil used was only a few atoms thick to reduce the density of the particles

3

The correct answer is **B** because:

- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- An α -particle is made of two protons and two neutrons, ${}_2^4\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- An γ -ray is usually emitted after alpha or beta decay to release excess energy from the nuclei
- The nucleon number is not changed in this case
 - The option that will result in the highest number of neutrons remaining in nucleus Y is therefore **B**

4

The correct answer is **C** because:

- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- An α -particle is made of two protons and two neutrons, ${}_2^4\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- The difference in mass between
 - ${}_{86}^{222}\text{Rn}$ and ${}_{82}^{206}\text{Pb}$ is $222 - 206 = 16$
- As beta decay does not decrease mass, the change must be due to alpha decay
 - Number of alpha particles released = $\frac{16}{4} = 4$

- The difference in atomic number
 - $^{222}_{86}\text{Rn}$ and $^{206}_{82}\text{Pb}$ is $86 - 82 = 4$
 - The four alpha particles released would have emitted $4 \times 2 = 8$ protons
- The number of beta particles released must, therefore, be 4 to increase the proton number to 82
 - Therefore, the total number of stages = 8

5

The correct answer is **C**

- An α -particle is made of two protons and two neutrons, $^4_2\alpha$
 - $^A_Z X \rightarrow ^{A-4}_{Z-2} Y + ^4_2\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle ($^0_{-1}\beta$) and antineutrino ($\bar{\nu}_e$)
 - $^A_Z X \rightarrow ^A_{Z+1} Y + ^0_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- The first stage emits an α -particle reducing the proton number by 2
- The next two stages both emit a β increasing the proton number by 2
 - The final nucleus must have the same number of protons and a different number of neutrons, therefore, option **C** is correct

6

The correct answer is **A** because:

- Alpha, beta and gamma have different properties as shown in the table below

PARTICLE	WHAT IS IT	CHARGE	RANGE IN AIR	PENETRATION	IONISATION
ALPHA (α)	2 PROTONS + 2 NEUTRONS	+2	FEW cm	STOPPED BY PAPER	HIGH
BETA (β^-)	ELECTRON	-1	FEW 10s OF cm	STOPPED BY FEW MM ALUMINIUM	MEDIUM
GAMMA (γ)	ELECTROMAGNETIC WAVE	0	INFINITE	REDUCED BY FEW MM LEAD	LOW

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- Diagram 1 shows the absorbance of the radiation by solids:
 - W is an alpha particle stopped by paper
 - X is a beta particle stopped by 1 mm of aluminium
 - Y is a gamma-ray reduced by lead
- Diagram 2 shows the radiations passing an electric field:
 - L is a beta-particle attracted to the positive terminal
 - M is a gamma-ray not deflected by the electric field
 - N is an alpha particle attracted to the negative terminal
- Diagram 3 shows:
 - R is a beta particle deflected by the magnetic field
 - P is an alpha particle deflected by the magnetic field
 - Q is a gamma-ray is not deflected by the magnetic field
- The correct option is therefore **A**

7

The correct answer is **D** because:

- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and antineutrino ($\bar{\nu}_e$)
 - $${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0\beta + \bar{\nu}_e$$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- This type of decay would result ${}_3^8\text{Li}$ in having one extra proton and one less neutron, so, the only correct option is **D**

8

- This is a gamma-ray not deflected by the electric field
 - N is an alpha particle attracted to the negative terminal
- Diagram 3 shows:
 - R is a beta particle deflected by the magnetic field
 - P is an alpha particle deflected by the magnetic field
 - Q is a gamma-ray is not deflected by the magnetic field
- The correct option is therefore **A**

7

The correct answer is **D** because:

- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- This type of decay would result ${}_3^8Li$ in having one extra proton and one less neutron, so, the only correct option is **D**

8

The correct answer is **B** because:

- The question states that after decay it is an isotope of thorium which has a proton number of 90
- An α -particle is made of two protons and two neutrons, ${}_2^4\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\alpha$
- The nucleon number is reduced by $232 - 228 = 4$, so, two neutrons and two protons are emitted
 - This would result in a proton number of 88, this is not correct as the final isotope should also have a proton number of 90
 - Therefore, two beta particles must be emitted to make up the final proton number
- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and antineutrino ($\bar{\nu}_e$)



- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
 - So, this would result in a proton number of 90

Therefore, one alpha particle and two beta particles are emitted

A is incorrect because in this option the final proton number would be 94

C is incorrect because in this option the final proton number would be 89

D is incorrect because in this option the final proton number would be 87

9

The correct answer is **D** because:

- The gold nucleus will cause a repulsive force on the alpha particle as they are all positively charged
- The gold nucleus would deflect the alpha particle away.
- The closer the particle the stronger the force and therefore the deflection
- The paths will not stay parallel due to a greater repulsive force on some of the particles

A is incorrect because the diagram shows an attraction towards the gold nucleus

B is incorrect because the diagram shows the paths not being crossed by the greater repulsive force, and an attraction of the alpha particle towards the nucleus

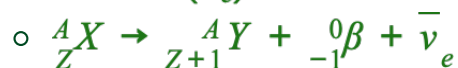
C is incorrect because the diagram shows the paths not being crossed by the greater repulsive force

10

The correct answer is **A** because:

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and

antineutrino ($\bar{\nu}_e$)



- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
 - The first stage emits an β -particle increasing the proton number by 1
 - The next stage emits an α -particle decreasing the proton number by 2
 - The final stage emits an β -particle increasing the proton number by 1
- The number of protons in Z are the same as W, but with a different number of neutrons
 - Therefore, W and Z are isotopes of each other

11

The correct answer is **C** because:

- An antiparticle is a subatomic particle having the same mass as one of the particles of ordinary matter but opposite electric charge and magnetic movement
- A positron is the antiparticle of the negatively charged electron
- They only differ in their charge
 - The positron has a +ve charge and the electron a -ve charge

A is incorrect because the mass of an electron and positron are the same

B is incorrect because the magnitude of charge is the same; the positron has a charge of +1 and the electron -1

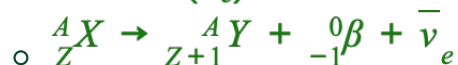
D is incorrect because the positron and electron both have the same spin of $\frac{1}{2}$

12

The correct answer is **D**

- An α -particle is made of two protons and two neutrons, ${}_2^4\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
 - Nucleus S goes from ${}_{84}^{134}\text{S}$ to ${}_{82}^{132}\text{X}$, so the first stage is α decay
- In β^- decay, a neutron decays to become a proton, β^- particle (${}_{-1}^0\beta$) and

antineutrino ($\bar{\nu}_e$)



- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
 - Nucleus X goes from ${}^{132}_{82}X$ to ${}^{130}_{84}T$, so two β^- particles must have been released

13

The correct answer is **C**

- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino ($\bar{\nu}_e$)
 - ${}^A_ZX \rightarrow {}^A_{Z+1}Y + {}^0_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
 - Nucleus X goes from ${}^{234}_{90}X$ and undergoes two β^- decays
 - The nucleon number will not change, so, it will still be 234
 - The proton number will increase by two, so, it will be 92
- Therefore, option **C** correct

14

The correct answer is **A**

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino $\bar{\nu}_e$
 - ${}^A_ZX \rightarrow {}^A_{Z+1}Y + {}^0_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- Astatine decays to form polonium emitting an ${}^0_{-1}\beta$ and ${}^4_2\alpha$

14

The correct answer is **A**

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino $\bar{\nu}_e$
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}^0_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
- Astatine decays to form polonium emitting an ${}^0_{-1}\beta$ and ${}^4_2\alpha$
 - Stage 1: ${}^{218}_{85}At \rightarrow {}^{218}_{86}X + {}^0_{-1}\beta$
 - Stage 2: ${}^{218}_{86}X \rightarrow {}^{214}_{84}Po + {}^4_2\alpha$

15

The correct answer is **C**

- An α -particle is made of two protons and two neutrons, ${}^4_2\alpha$
 - ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}^4_2\alpha$
- The nucleon number is reduced by 4, emitting two neutrons and 2 protons
- In β^- decay, a neutron decays to become a proton, β^- particle (${}^0_{-1}\beta$) and antineutrino $\bar{\nu}_e$
 - ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}^0_{-1}\beta + \bar{\nu}_e$
- The mass is not changed, but the proton number increases by one and the number of neutrons decrease by one
 - If the first decay emits an β^- -particle, it will increase the proton number by 1
 - If the second decay emits an α -particle, it will decrease the proton number by 2
 - If the third stage emits an β^- -particle, it will increase the proton number by 1
- The number of protons will remain the same after three decays, 1 α decay and 2 β^- decays

- This makes them isotopes of each other

